

INFORMATION TECHNOLOGY INSTITUTE

LAB 10

Noise Simulation





Contents

PART 1: LPF AC Noise Analysis	4
Schematic	4
AC analysis.....	4
Hand analysis.....	5
Parametric sweep	6
Comment.....	6
PART 2: LPF Transient Noise Analysis	7
TSTEP = TAU/100	7
noise output waveform	7
.....	7
TSTEP = TAU/10	8
20 run	9
Value per iteration.....	9
Average	9
Comparison	9
PART 3: 5T OTA AC Noise Analysis	10
Schematic	10
Report output thermal noise vs frequency	10
Hand analysis.....	11
Total output noise (thermal + flicker) vs frequency	12
Noise Summary.....	12
Calculating RMS	12
PART 4: 5T OTA Transient Noise Analysis.....	13
Plot input and output	13

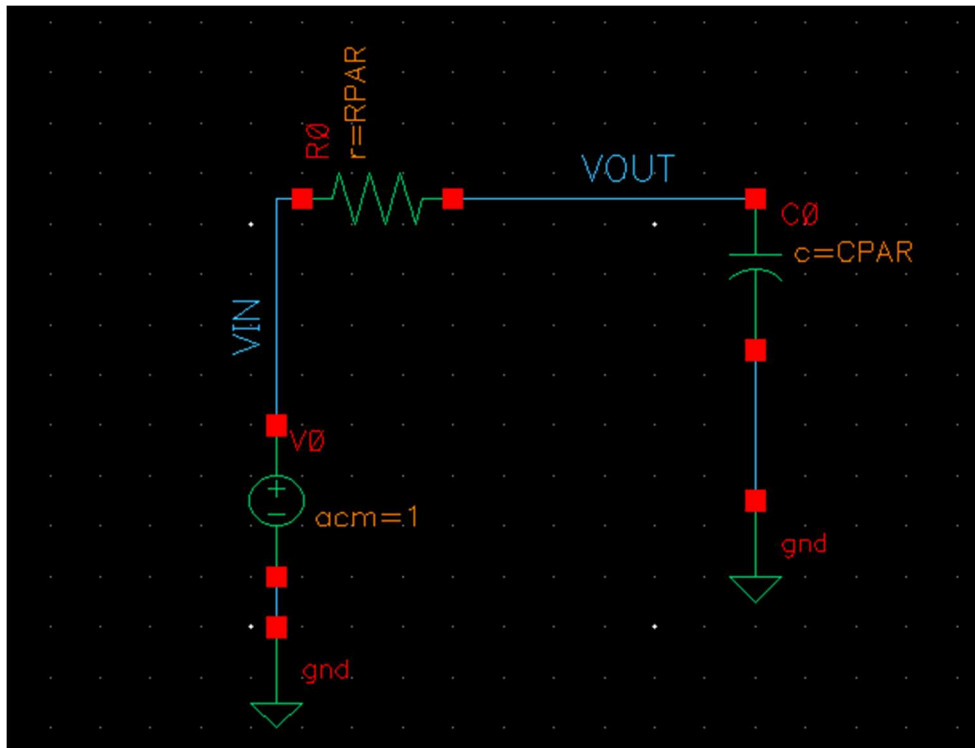


Transient Noise Analysis	13
run 20 simulation runs.....	14
rms per iteration.....	14
Average 20 run rms	14



PART 1: LPF AC Noise Analysis

Schematic



AC analysis





Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:lab_09_noise_rc:1	RMS noise	64.32u			
lab10:lab_09_noise_rc:1	bw	158.8M			
lab10:lab_09_noise_rc:1	vout				

Hand analysis

$$R = 1 \text{ k}\Omega, \quad C = 1 \text{ pF}, \quad T = 300.15 \text{ K}$$

$$v_{n,out}(f) = \frac{\sqrt{4kTR}}{\sqrt{1+(2\pi fR)^2}}$$

$$v_{n,out}(0) = \sqrt{4kTR} = 4.071 \text{ nV}/\sqrt{\text{Hz}}$$

$$f_{3dB} = \frac{1}{2\pi RC} = 159.15 \text{ MHz}$$

$$B_n = \frac{\pi}{2} f_{3dB} = \frac{1}{4RC} = 250 \text{ MHz}$$

$$v_{n,rms} = \sqrt{4kTRB_n} = \sqrt{\frac{kT}{C}} = 64.37 \text{ }\mu\text{V}$$

Parameter	Hand Analysis	Simulation	Error
Noise density	4.071 nV/ $\sqrt{\text{Hz}}$	4.071 nV/ $\sqrt{\text{Hz}}$	0.001%
Bandwidth	159.15 MHz	158.8 MHz	0.223%
RMS output noise	64.37 μV	64.32 μV	0.084%



Parametric sweep



Comment

As the resistance increases the noise increase psd as thermal noise is $4kTR$ but the bandwidth decreases as the $BW = 1/2\pi * R * C$.

Point	Test	Output	Nominal	Spec	Weight	Pass/Fail
Parameters: RPAR=1k						
1	lab10:lab_09_noise_rc:1	RMS noise	64.32u			
1	lab10:lab_09_noise_rc:1	bw	158.8M			
1	lab10:lab_09_noise_rc:1	vout				
Parameters: RPAR=10k						
2	lab10:lab_09_noise_rc:1	RMS noise	64.62u			
2	lab10:lab_09_noise_rc:1	bw	15.88M			
2	lab10:lab_09_noise_rc:1	vout				
Parameters: RPAR=100k						
3	lab10:lab_09_noise_rc:1	RMS noise	64.65u			
3	lab10:lab_09_noise_rc:1	bw	1.588M			
3	lab10:lab_09_noise_rc:1	vout				
Parameters: RPAR=1M						
4	lab10:lab_09_noise_rc:1	RMS noise	64.66u			
4	lab10:lab_09_noise_rc:1	bw	158.8k			
4	lab10:lab_09_noise_rc:1	vout				

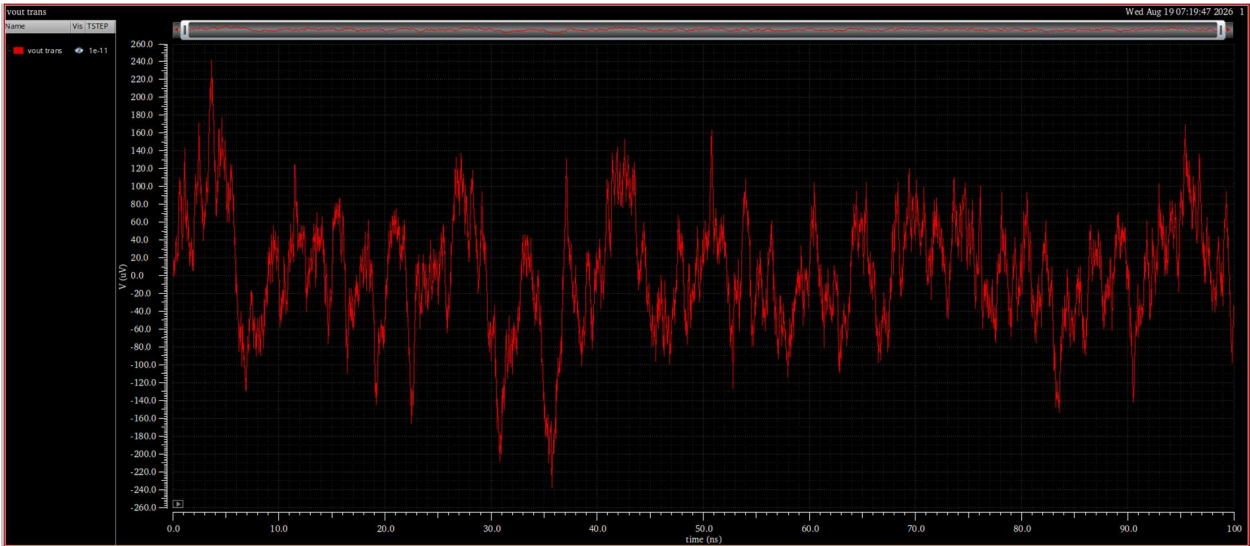
but the RMS value should stay the same as the it is $v_{rms} = \sqrt{\frac{kT}{C}}$.



PART 2: LPF Transient Noise Analysis

$TSTEP = \tau / 100$

noise output waveform



Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:lab_09_noise_rc:1	RMS noise	64.32u			
lab10:lab_09_noise_rc:1	bw	158.8M			
lab10:lab_09_noise_rc:1	vout				

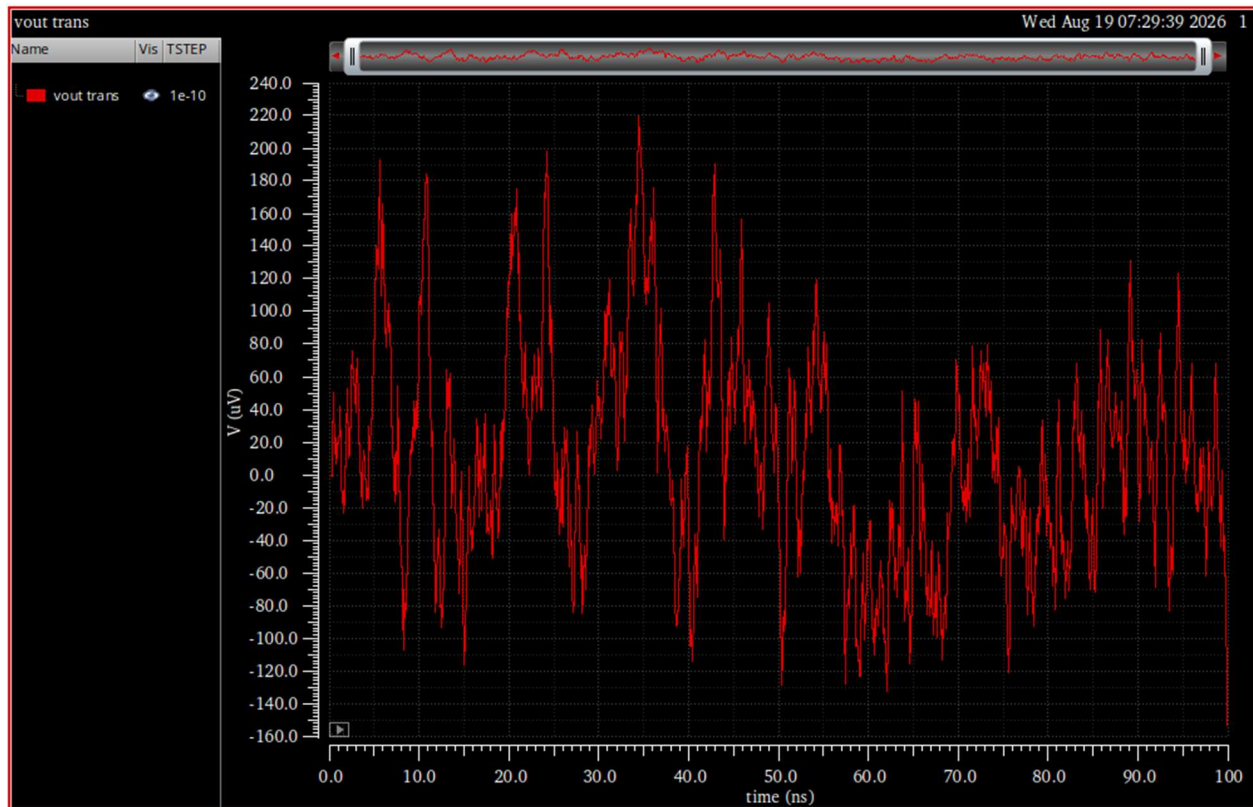
Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:lab_09_noise_rc:1	vout trans				
lab10:lab_09_noise_rc:1	rms	63.53u			

transient	Noise analysis
63.53	64.32

It is a bit lower because there is a time step doesn't tend to zero so the points are finite



$$TSTEP = \tau/10$$



Test	Output	Nominal	Spec	Weight	Pass/Fail
lab10:lab_09_noise_rc:1	vout trans				
lab10:lab_09_noise_rc:1	rms	66.57u			

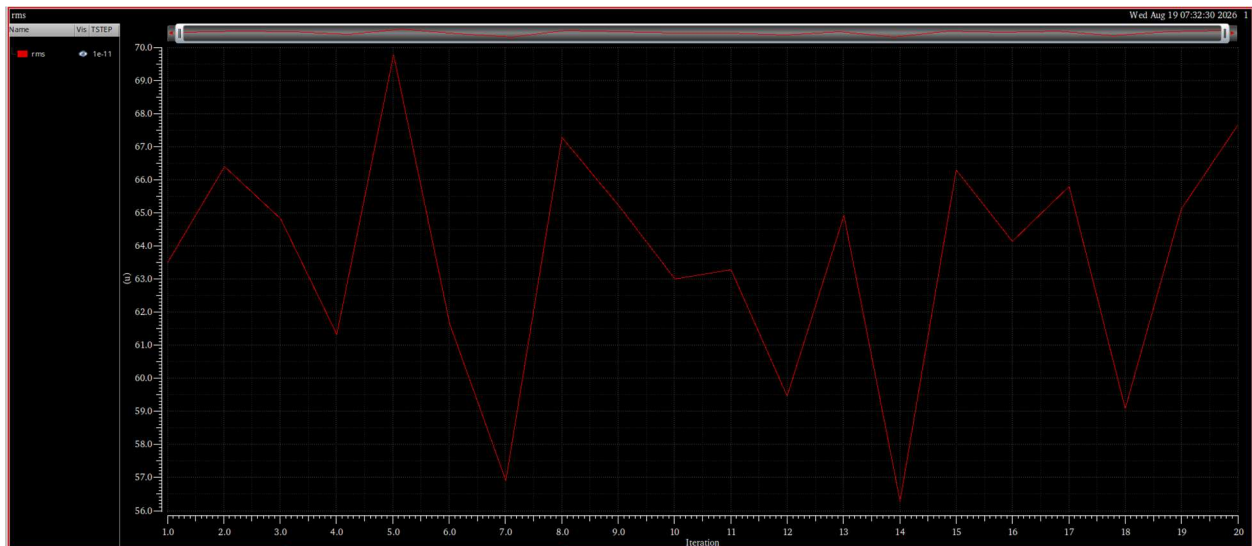
It increased as it only ran one random wave form

However if both were the same wave form it should have decreased as the F_{max} increased but this is negligible because F_{max} is larger than the BW of the RC circuit.

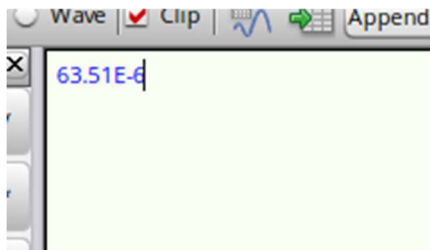


20 run

Value per iteration



Average



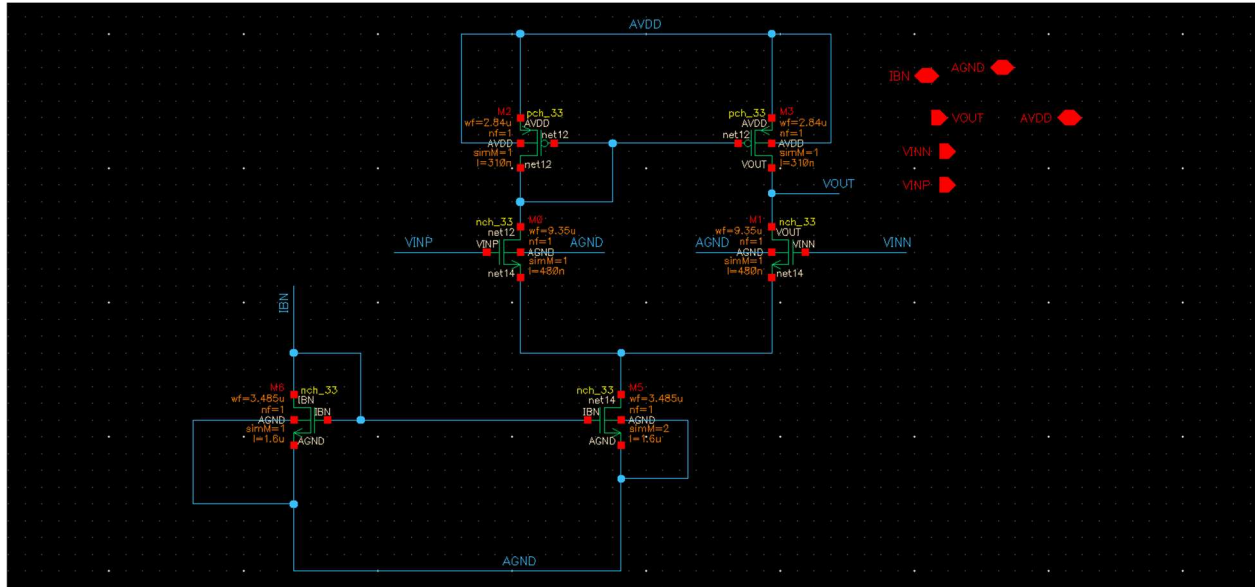
Comparison

Noise analysis	Single run	20 runs
64.32	63.53	63.51

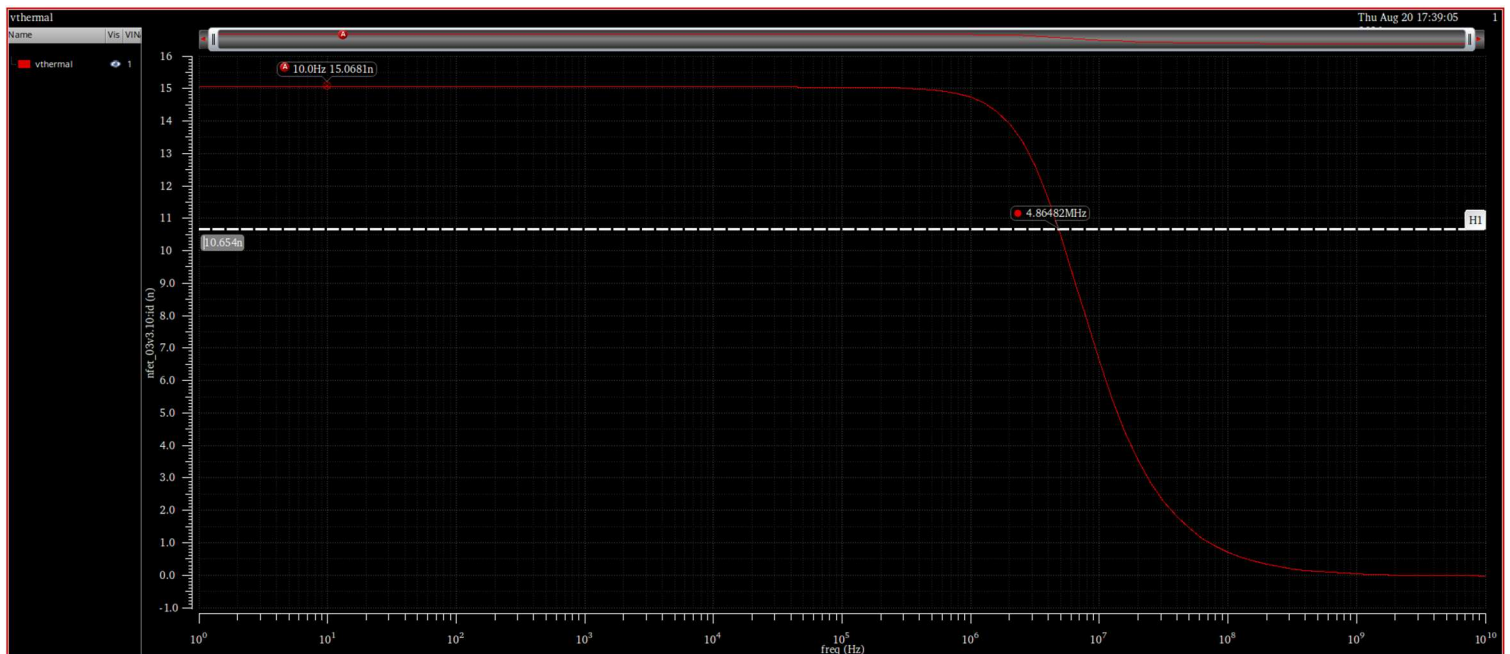


PART 3: 5T OTA AC Noise Analysis

Schematic



Report output thermal noise vs frequency





bandwidth(sqrt((pv('11.M0.m0' "id" ?r... x	
VINAC	bandwidt.. "low")
1 1.000	4.853E6

Hand analysis

$$g_{mn} = \frac{162.5 + 152.9}{2} = 157.7 \mu S$$

$$g_{mp} = \frac{80.52 + 74.93}{2} = 77.725 \mu S$$

Assume $\gamma=3/2$

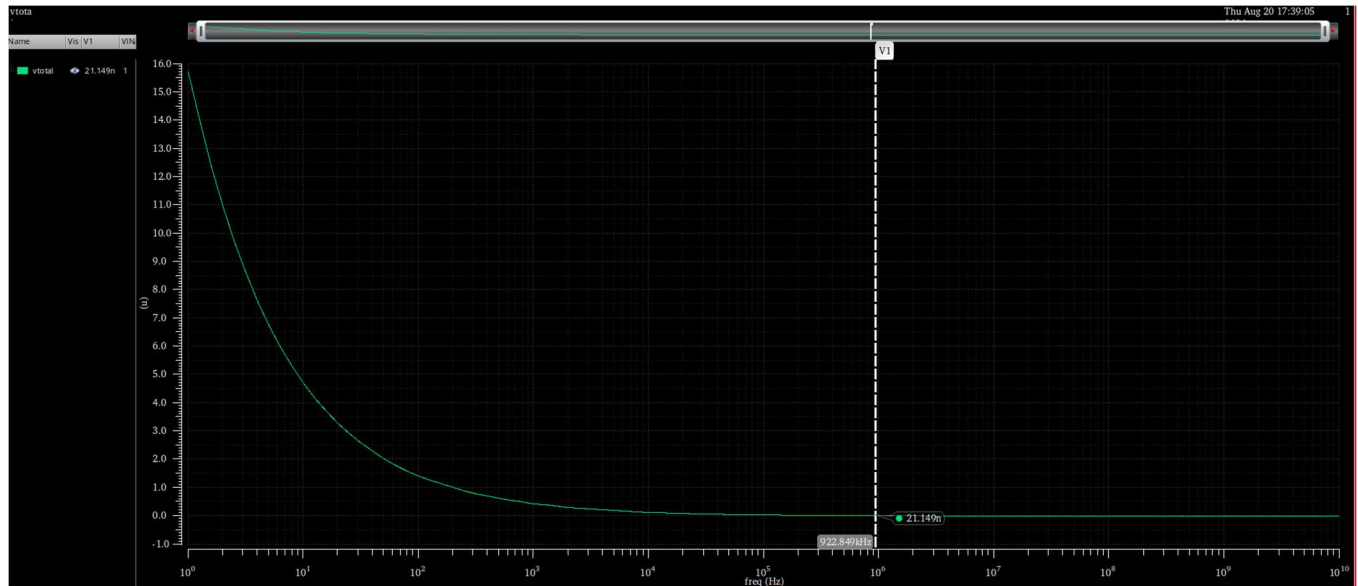
$$v_{n,th} = \sqrt{\frac{8kT\gamma}{g_{mn}} \left(1 + \frac{g_{mp}}{g_{mn}}\right)} \quad v_{n,th} = 14.46 \text{ nV}/\sqrt{\text{Hz}}$$

$$f_{BW} \approx \frac{g_{mn}}{2\pi C_L}, f_{BW} = \frac{157.7 \times 10^{-6}}{2\pi(5 \times 10^{-12})} = 5.02 \text{ MHz}$$

Parameter	Hand Analysis	Simulation
Thermal-noise density	14.46 nV/ $\sqrt{\text{Hz}}$	15.068 nV/ $\sqrt{\text{Hz}}$
Bandwidth	5.02 MHz	4.853 MHz



Total output noise (thermal + flicker) vs frequency



FC is when the total is equal to the $\sqrt{2} v_{(noisetherma)}$

Fc is around 922.85MHz

Noise Summary

Calculating RMS

$$B_n = \frac{\pi}{2} f_{BW}$$

$$B_n = \frac{\pi}{2} (5.02 \text{ MHz}) = 7.885 \text{ MHz}$$

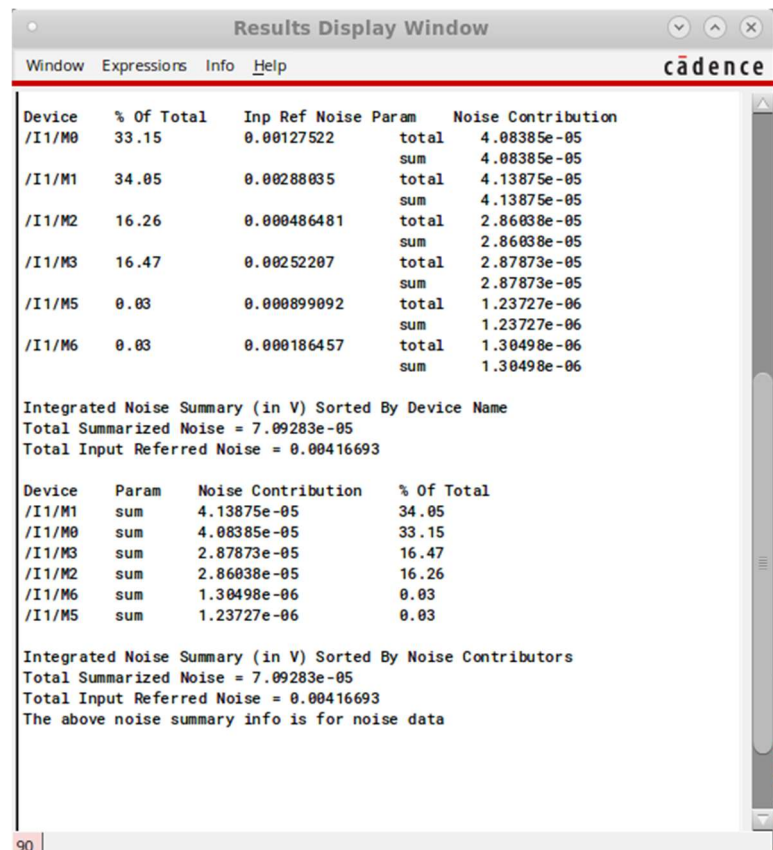
$$v_{n,th} = 14.46 \text{ nV}/\sqrt{\text{Hz}}$$

$$v_{n,rms} = v_{n,th} \sqrt{B_n}$$

$$v_{n,rms} = (14.46 * 10^{-9}) \sqrt{7.885 * 10^6}$$

$$v_{n,rms} = 40.62 \mu\text{V}$$

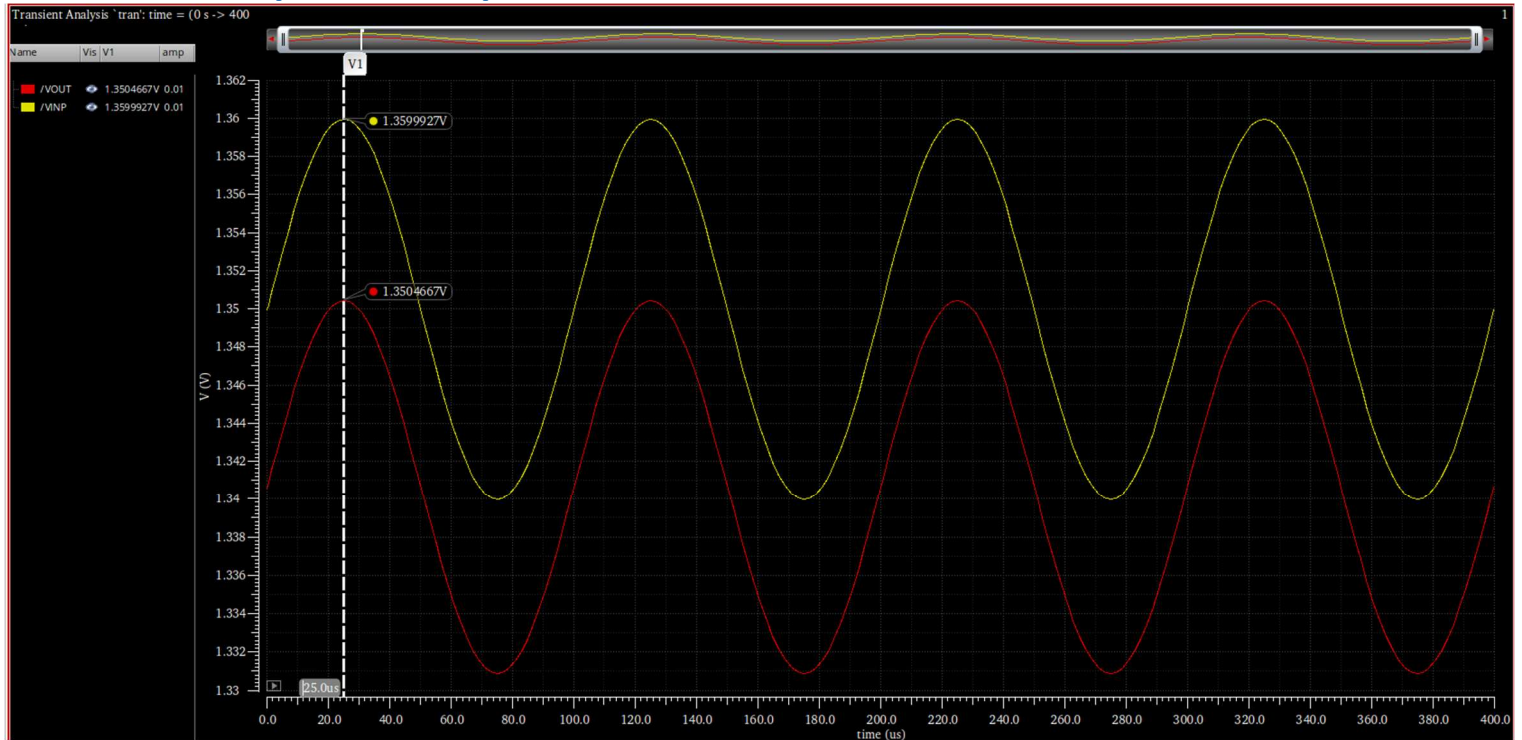
Bandwidth and density calculated previously



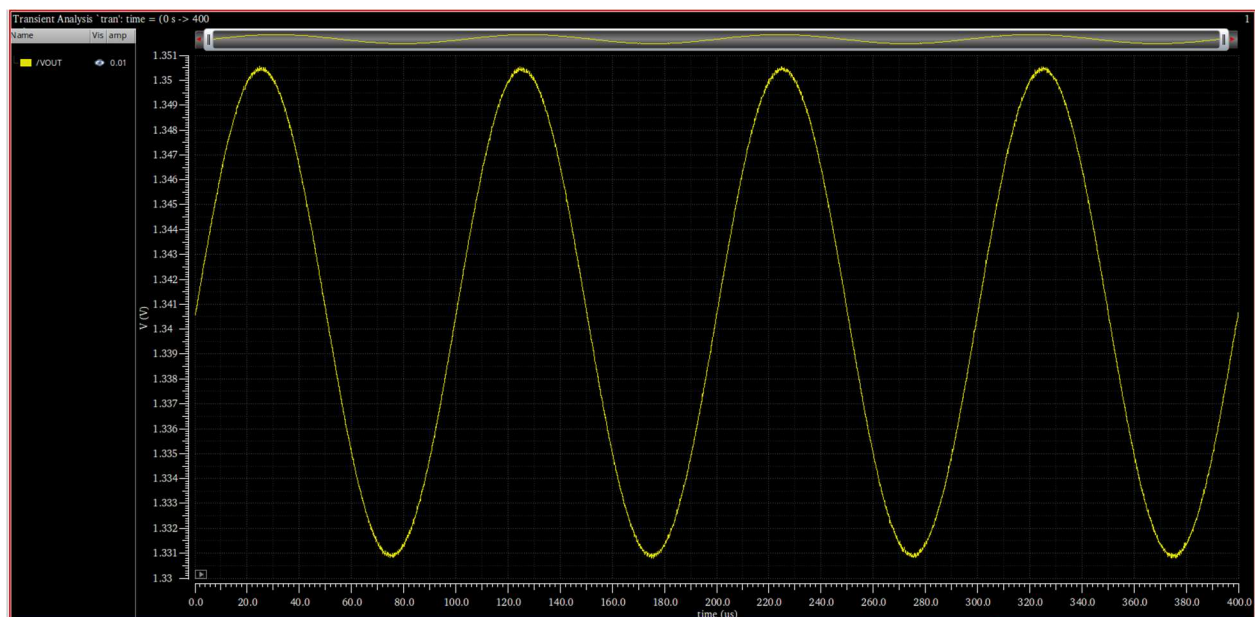


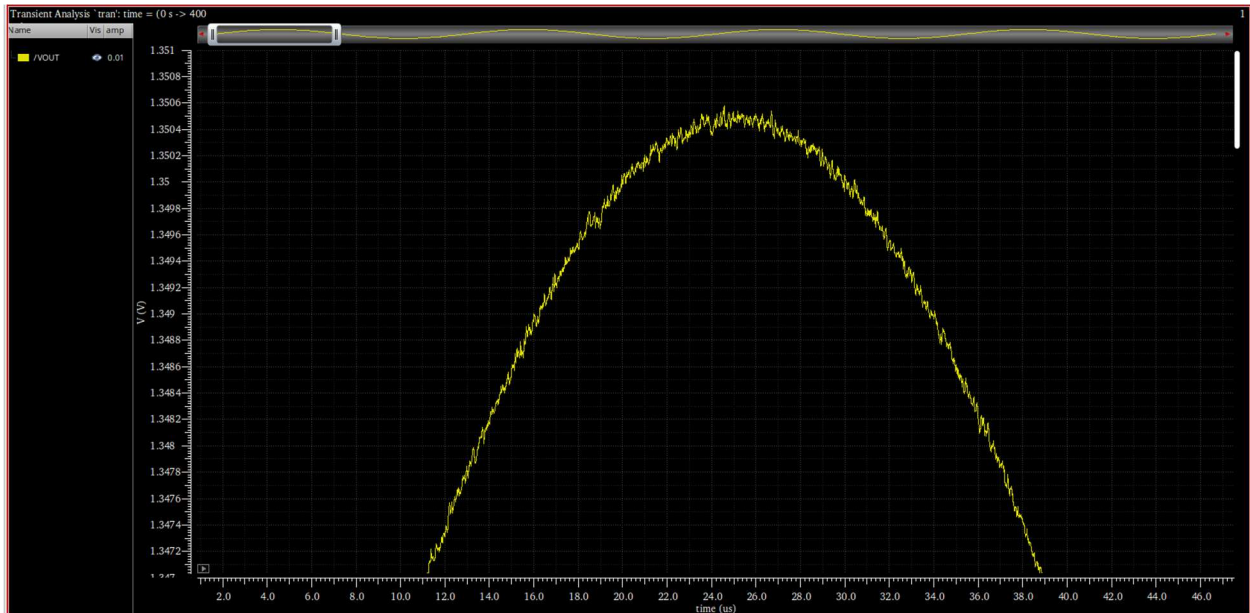
PART 4: 5T OTA Transient Noise Analysis

Plot input and output



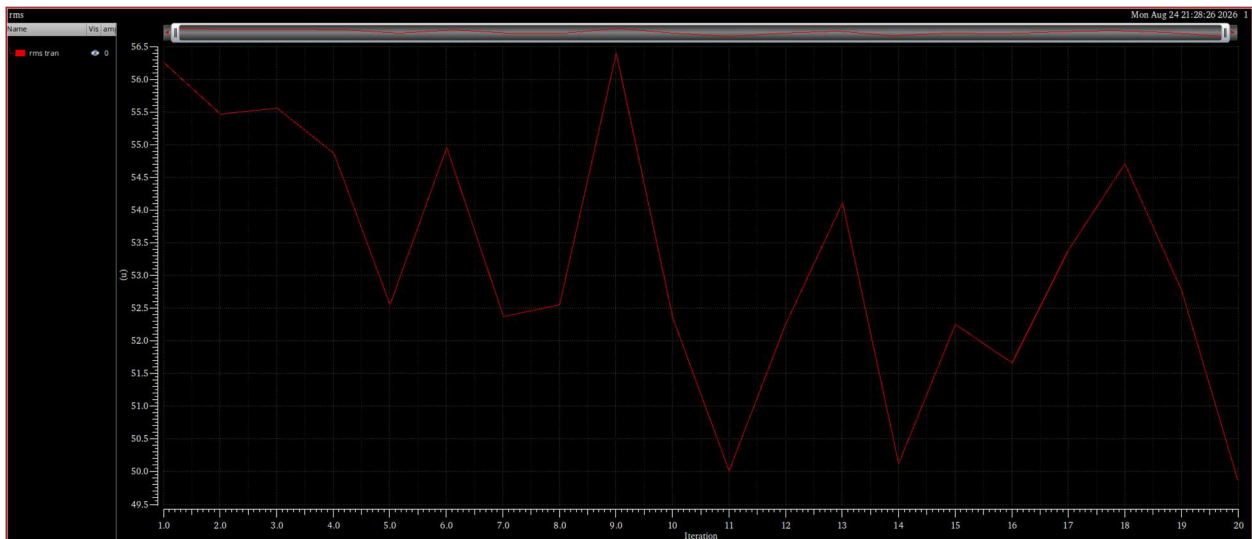
Transient Noise Analysis





run 20 simulation runs.

rms per iteration



Average 20 run rms

average(value(rms(VT("/VOUT") - aver... ×	
Expression	Value
1 average(value(rms(VT("/VOUT") - average(VT("/VOUT")))) "amp" 0))	53.24E-6

Expression used : $\text{rms}((\text{VT}("/\text{VOUT}") - \text{average}(\text{VT}("/\text{VOUT}"))))$ and $\text{VIN} = 0$ ac



calculated using ac noise analysis but with from frequency 2.5k till 50MHz

lab10:TB_ota:1	rmsNoise(2500 50000000)	56.85u		
----------------	-------------------------	--------	--	--

Transient avg	ac noise 2.5k:50MHz	ac noise 10:10GHz
53.24uV	56.85uV	70.9uV